

Chapter 5: Parallel and Intersecting Lines

Worksheet 5: Parallel and Intersecting Lines

Learning Objective: Identify angle relationships formed when two lines intersect, and when a transversal cuts a pair of parallel lines (vertically opposite, linear pair, corresponding, alternate and co-interior angles).

Worked Example:

Two lines intersect, forming four angles. If one angle is 70° , find the other three.

Solution: The vertically opposite angle is also 70° . The other two angles (each forming a linear pair with 70°) are $180 - 70 = 110^\circ$ each.

Questions (1–20)

1. What is the sum of angles on a straight line (a linear pair)?
2. If two lines intersect and one angle is 65° , what is its vertically opposite angle?
3. Two parallel lines are cut by a transversal. What is the relationship between corresponding angles?
4. Define alternate interior angles in one sentence.
5. Two co-interior angles (between parallel lines, same side of transversal) – if one is 110° , find the other.
6. Two lines intersect, forming an angle of 48° . Find the measures of the other three angles.

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7. A transversal cuts two parallel lines, and one corresponding angle is 75° . Find its corresponding angle on the other line.

 8. Lines AB and CD intersect at O. If angle $AOC = 130^\circ$, find angle BOD and angle AOD.

 9. Two parallel lines are cut by a transversal, and one alternate angle is 105° . Find its alternate pair.

 10. If two co-interior angles formed by a transversal are equal, what must each angle measure?

 11. A transversal cuts two parallel lines, making one angle 62° . Find all the other angles formed at both intersection points.

 12. Two lines intersect, forming 4 angles, two of which are right angles. What can you say about the two lines?

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13. A road crosses two parallel railway tracks. If the angle between the road and one track is 55° , what is the angle between the road and the other track?
14. Two co-interior angles are $3x$ and $2x$. Find x and the measure of each angle.
15. Three lines intersect at a single point, forming six angles. If one angle is 50° and an adjacent one is 70° , find the remaining four angles (using vertically opposite pairs).
16. A ladder leans against a wall, making a 65° angle with the ground. If a shelf parallel to the ground also meets the ladder, what angle does the ladder make with the shelf?
17. Two parallel lines are cut by a transversal. One pair of alternate angles are $(2x + 10)^\circ$ and $(3x - 15)^\circ$. Find x and each angle.
18. Two parallel railway tracks are crossed by a road. One angle formed is 38° more than its co-interior angle. Find both angles.

19. Look at a window grill, ladder, or railway track near your home. Identify a pair of parallel lines and a transversal (if any), and estimate one angle formed.
20. Find an example of intersecting lines in your classroom (such as a door frame). Estimate the angles formed at the intersection.

Reflection Question: Why do you think railway tracks are built parallel to each other? What would happen if they were not parallel?

Real-Life Application Question: Find an object at home or school showing two lines crossing (scissors, an X-shaped gate, folded fabric) and describe the angles you see.